



Society of Petroleum Engineers

SPE-230156-MS

Enhancing Metering System Oversight and Assurance Through PMMS Real-Time Dashboards and Integrated Tools

A. Hinai, F. Kalbani, Y. Naumani, and M. Balushi, Petroleum Development Oman, Sultanate of Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230156-MS](https://doi.org/10.2118/230156-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

The Production Metering Management System (PMMS) is a digital solution developed to support the Production Engineering community within Petroleum Development Oman (PDO) by enabling reliable, centralized monitoring of the health and performance of production metering systems. With over 6,000 metering devices distributed across four classification categories: Custody Transfer, Allocation, Well Testing, and Environmental PDO faced significant challenges due to fragmented data sources, inconsistent workflows, and the absence of a unified monitoring platform. PMMS addresses these challenges by acting as a single-window interface that integrates multiple corporate databases and systems, delivering correlated and contextualized data to enhance operational visibility, measurement assurance, and lifecycle management.

This paper presents the architecture, development, and implementation of PMMS, emphasizing its role in unifying scattered data across databases, documents, drawings, procedures, and standards. The solution not only streamlines metering workflows but also supports proactive maintenance and strategic decision-making. The innovative approach outlined in this paper exemplifies industry best practices for managing safety-critical systems and achieving operational excellence through digital transformation and integrity assurance.

Introduction

Metering systems are a critical component in the oil and gas industry, underpinning accurate production measurement and ensuring operational efficiency across multiple domains. Within Petroleum Development Oman (PDO), over 6,000 metering devices are deployed, categorized into four primary classes: Custody Transfer, Allocation, Well Testing, and Environmental. These devices operate across diverse field stations and are managed by specialized teams, each focusing on distinct operational objectives. Despite their essential role, the absence of a centralized platform has posed significant challenges in governing, assuring, and continuously monitoring the overall condition and performance of the metering infrastructure.

To overcome these challenges, the Production Measurement Management System (PMMS) was developed as an integrated digital solution tailored to meet the needs of the Production Engineering community within PDO, as illustrated in [figure 1](#). PMMS offers a unified interface that consolidates data

from multiple corporate databases, technical documents, engineering drawings, operational procedures, and standards, which were previously fragmented and dispersed. By aggregating and contextualizing this data, PMMS provides a comprehensive and coherent view of metering assets, enhancing operational visibility and enabling more efficient and informed decision-making. This paper details the conceptualization, architecture, and implementation of PMMS, highlighting its strategic role in streamlining production measurement workflows, supporting proactive maintenance, and elevating measurement assurance to industry best practices. Furthermore, it explores the technical and organizational challenges faced and the resulting benefits in driving operational excellence and integrity management.

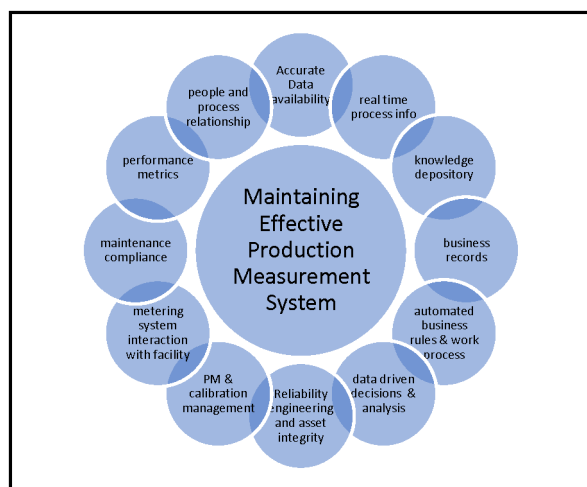


Figure 1—Meter Management Solution

Methodology

The methodology employed begins with compiling a comprehensive database of 10,000 production metering devices from all assets, referencing Process Engineering Flow Schematic (PEFS) drawings. This is subsequently followed by the identification and classification of the metering devices based on the four metering classes, their respective functions, and associated uncertainty levels. Furthermore, all the metering systems are verified by a system engineer within the Distributed Control System (DCS) to ensure that the configuration of the metering devices complies with established standards and requirements. The Production Measurement Management System (PMMS) is integrated with the Distributed Control System (DCS) via the Process Information (PI) data historian. The metering devices are configured within PMMS and mapped to PI tags, thereby creating a synchronized database essential for establishing critical information, as illustrated in [figure 2](#). This project is built on Agile methodology, enabling iterative progress, continuous feedback, and adaptive planning throughout the development lifecycle. This helps the users to adapt easily to evolving requirements and maintain strong team collaboration.

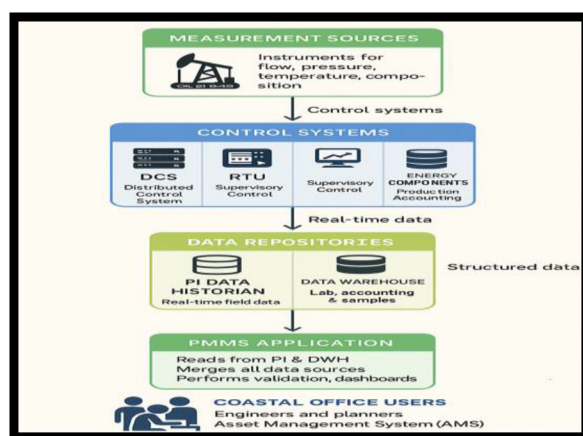


Figure 2—Data Flow

PMMS Functionality

PMMS is a specialized software platform designed to streamline and enhance the management of production metering systems across oil and gas operations. Its core functionalities and applications include:

Plant Hierarchy

This level corresponds to the physical locations within the organization's infrastructure, encompassing Directorates, Clusters, and Stations, which constitute the primary operational divisions. Users can access instruments based on their respective locations via a hierarchical tree view system. By expanding nodes within this hierarchy, users can progressively navigate through Directorates, Clusters, and Stations. Selecting the relevant nodes grants access to schematics associated with the chosen location. This structured hierarchy facilitates intuitive navigation and visualization of instrument distribution across various organizational levels, thereby enhancing user comprehension and accessibility to relevant information.

Meter Performance Dashboard

PMMS facilitates to address the metering challenges to design the metering system into these below mentioned dashboards to monitor the health and performance of metering system across the organization.

Class 1 & 2 Allocation and fiscal metering dashboard

Performance metrics for Main Oil Line (MOL) Class 1 and Class 2 meters are critical to ensuring accurate measurement and refining processes. These metrics represent real-time data for each stream, including flow rates, pressure, temperature, and other pertinent parameters. Interactive charts, graphs, and gauges are utilized to visualize key parameters, facilitating straightforward interpretation by operators.

These performance metrics establish a comprehensive framework for evaluating the effectiveness and reliability of MOL Fiscal and Allocation meters throughout Petroleum Development Oman (PDO). Continuous monitoring and evaluation of these metrics are imperative for maintaining measurement accuracy and optimizing operational efficiency.

The dashboard also incorporates operating envelope performance analysis, offering insights into the extent to which instruments operate within their designated limits. Users can view the operating envelope boundaries and assess how actual performance metrics compare to these thresholds. Deviations from the operating envelope may signal potential issues or areas requiring improvement, thereby prompting further investigation and corrective action.

The Production Measurement Management System (PMMS) enhances this process by employing real-time color coding to indicate meter performance status. An intuitive color scheme is utilized to differentiate between normal operation, warning conditions, and critical alarms. Additionally, the PMMS integrates an action tracker model to document actions taken and monitor progress directly within the dashboard. This feature facilitates regular review sessions among relevant stakeholders, enabling monthly assessments to verify the performance and condition of individual meters and drive improvements in meter-related key performance indicators (KPIs).

The dashboard, as illustrated in Figure 3, presents real-time data for each stream's flow, pressure, and temperature. Additionally, FY associated with the standard calculations configured in the Distributed Control System (DCS) is continuously monitored and reviewed to ensure compliance with international standards. These parameters can be accessed via the Trend feature, which enables real-time monitoring of meter performance within the defined operating envelope. Furthermore, the exclamation mark icon (!) displayed above each meter with a red status indicates the presence of an issue, which may include the meter being out of service, a communication fault, or other related problems.

Directorate	Area	MOL Station	Status	F1	WC	P1	T1	FY	SAP Compliance
Directorate A	Area 1	Station A1							
Directorate B	Area 2	Station B2							
Directorate C	Area 3	Station C3							
Directorate D	Area 4	Station D4							
Directorate E	Area 5	Station E5							
	Area 6	Station E6 - 1							
		Station E6 - 2							
	Area 7	Station E7 - 1							
		Station E7 - 2							
		Station E8 - 3							
		Station E9 - 4							
		Station E7 - 5							
	Area 8	Station E7 - 6							
		Station E8 - 7							
		Station E8 - 8							

Figure 3—MOL Meter Performance Metrics

Meter Status

The performance of each meter is represented through a color-coded system that reflects its operational status. This visual indicator allows relevant stakeholders to promptly identify, investigate, and address any underlying issues. The legend below explains the color codes used to denote each meter's performance relative to its defined operating envelope, as illustrated in figure 4. Four color codes are incorporated into the dashboard, based on the percentage of data points falling within the operating envelope, and are defined as follows:

1. Green: 80% of the time the data is inside the defined operating envelope
2. Yellow: $\geq 50\%$ & $< 80\%$ of the time the data is inside the defined operating envelope
3. Orange: $\geq 20\%$ & $< 50\%$ of the time the data is inside the defined operating envelope
4. Red: $< 20\%$ of the time the data is inside the defined operating envelope

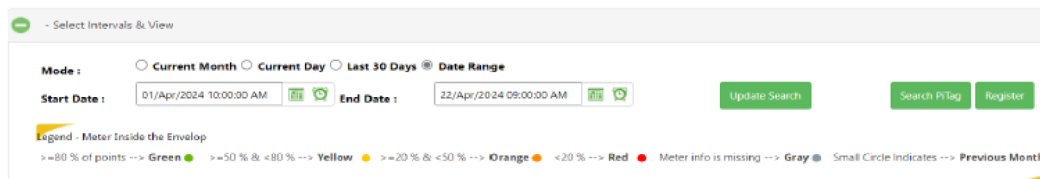


Figure 4—Status color coding

This color-coding system enhances the clarity and efficiency of monitoring each meter's performance by providing immediate visual cues. It acts as an early warning mechanism, enabling stakeholders to promptly identify potential issues and undertake necessary verification and investigation of the meter's behavior before problems escalate.

Integration with notification

Integrate the dashboard with notification systems to send alerts via emails, SMS, or other communication channels. Ensure that concerned teams receive timely notifications. By implementing these features, the MOL Water Cut Meter Live Operating Envelope Performance Analysis Dashboard provides a comprehensive tool for monitoring and managing water cut levels, ensuring adherence to operating envelopes, and facilitating timely intervention when deviations occur, as illustrated in [figure 5](#).

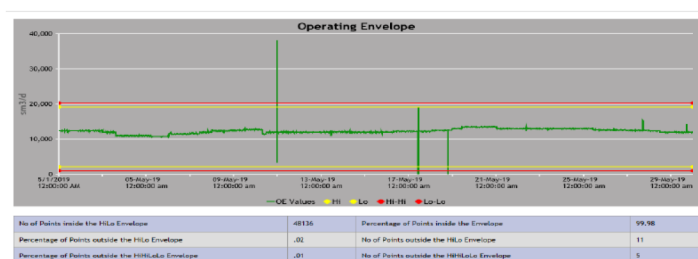


Figure 5—Real Time operating Envelop

Meter Performance Assurance

The Class 1 and Class 2 dashboards convene monthly meetings with relevant stakeholders to ensure a thorough review of the action tracker for registered tasks, assess the performance and condition of the metering system, and address any challenges along with corresponding countermeasures. These meetings also ensure that appropriate maintenance activities are carried out promptly and that historical data is accurately uploaded into the system. This regular coordination significantly contributes to a clearer understanding of the metering system's requirements, reduction in meter failures, and the sustained reliability and accuracy of the meters. Similar procedures are applied to all other dashboards, which also hold monthly meetings to maintain the ongoing reliability and accuracy of the metering system.

Explanation of the Meter Factor Envelope control chart in [figure 7](#):

1. Meter factor monthly proving values (Green): These points represent the actual monthly measurements of the meter factor, showing how the meter is performing over time
2. Meter factor deviation limit (Yellow): The yellow lines indicate the acceptable range of variation around the meter factor values. They are set as the meter factor value plus 0.1 and minus 0.1, defining a tolerance band within which the meter factor is considered stable.
3. +3/-3 sigma limits (Red): The red lines represent the control limits set at plus and minus three standard deviations (sigma) from the mean meter factor. These limits help identify when the meter factor values are statistically unlikely and may indicate an out-of-control condition.
4. Calculation of Deviation limits: The meter factor deviation limits (Yellow Lines) are derived based on the standard deviation of the meter factor values, providing a threshold that accounts for normal variability in measurements.

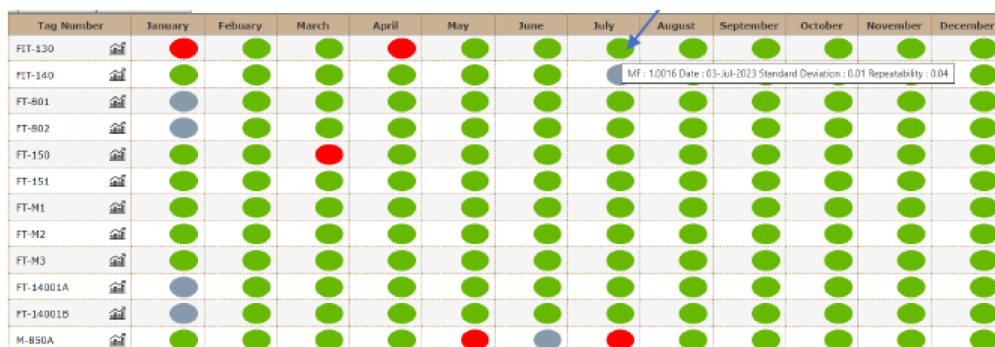


Figure 6—Fiscal Meter Proving Assurance Dashboard

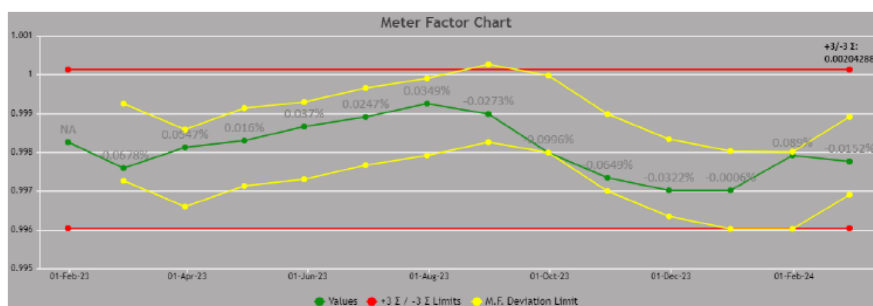


Figure 7—Meter Factor Control Chart

Expanded Scope of PMMS Functionality

The Production Metering Management System (PMMS) is designed not only to monitor and manage the performance of Main Oil Line (MOL) production meters but also to serve as a versatile platform supporting a broad range of metering applications across the production network. Its flexible configuration capabilities allow it to manage and monitor meters used in various operational domains, ensuring consistency, reliability, and visibility across the entire production system.

Specifically, PMMS dashboard supports meter configuration and performance tracking for:

- Gas Network Stations – Ensures accurate measurement and monitoring of gas distribution across different stations in the gas supply network.
- Flare Metering – Facilitates compliance with environmental regulations by tracking flare volumes and detecting flare limit exceedances.
- Product Quality Monitoring (BS&W and Gas Chromatography) – Supports the validation of product quality by integrating with Basic Sediment & Water (BS&W) sensors and Gas Chromatographs (GCs).
- Steam Applications – Enables the monitoring of steam flow meters used in thermal processes and enhanced oil recovery.
- Gas Lift Systems – Assists in optimizing gas lift operations by tracking injection volumes and performance across wells.
- Gas Injection Systems – Monitors gas reinjection activities to support reservoir pressure maintenance and enhanced recovery.
- Water Disposal Systems – Ensures proper metering of produced water disposal volumes for environmental and regulatory compliance.
- Wet Gas Metering – Accommodates the complexities of wet gas flow measurement, providing tools for monitoring and performance assurance.

- By extending its capabilities beyond conventional oil metering, PMMS supports a holistic approach to production measurement management, ensuring consistency, integrity, and efficiency across various production processes and facilities.

Live Production Monitoring Dashboard

This feature offers predictive insights into allocated production figures for all streams, alongside real time access to actual data. Acting as a centralized hub, this dashboard integrates data from various corporate database, streamlining production process and proving correlated and contextualized information for investigation daily production drops, as illustrated in [figure 8](#). It enables timely detection of communication failures for critical streams, including export meter, EN-RAF, and AP & LP flare meters.

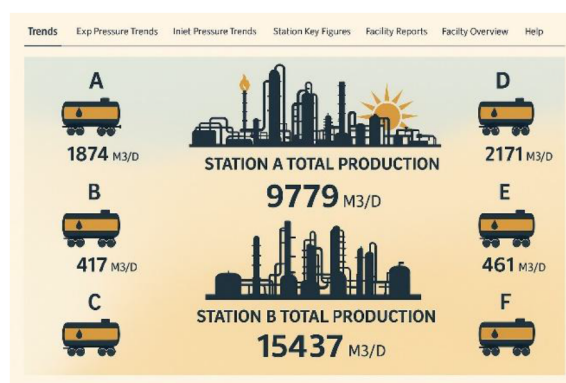


Figure 8—Live Production Data

The dashboard serves as a crucial tool for daily production oversight and investigation of production drops. Here's an overview of its key features:

- Provides a comprehensive production overview and facilities analysis of production drops
- Presents allocated production trends and trains trends sourced from energy component for detailed analysis
- Calculates actual live production figures by aggregating data from multiple corporate databases
- Displays total flare and water injection data prominently on the homepage for quick reference
- Includes oil and pressure nodes data for deeper insights and analysis into production trends and performance

Provides schematic diagrams of station facilities with real time values, offering a visual representation of production assets and their status.

Well Test Details

The Well Testing module within PMMS enables reliable monitoring of well test performance for test separators, Multiphase Flow Meters (MPFM), and Coriolis meters with water cut measurement across PDO operations. The data obtained from well tests supports the team in evaluating total gross production and water cut, which are essential for calculating net oil output. Additionally, the module offers clear visualization of well performance and trends for key parameters such as Gas Volume Fraction (GVF), oil, gas, Basic Sediment and Water (BS&W), pressure, and validation status, thereby enhancing the overall understanding of well behavior and performance, as illustrated in [Figure 9](#).



PMMS features to build Auto Verification System is a proposed solution for verifying metering data & accuracy in metering data is crucial for operations as it forms the foundation for critical decision-making processes. However, the manual verification of data collected from various stations is prone to human errors, leading to potential discrepancies and compromised reliability. The Auto Verification System aims to automate the metering verification process, ensuring the accuracy & reliability of data collected from various stations. By implementing this system, improves operational efficiency, enhance data accuracy, and facilitate sound decision-making.

- Enhanced Data Accuracy
- Improved Operational Efficiency
- Timely Error Detection and Resolution
- Regulatory Compliance
- Real-time Monitoring and Reporting
- Alert notification for high deviation

The auto-verification feature is designed to compare two output values: the standard value recorded from the DCS to the PI system and the corresponding value calculated within the PMMS platform, based on AGA 8 and ISO 6976 standards. The application compares these values and allows for an acceptable deviation within $\pm 2\%$. Any discrepancy exceeding this threshold triggers an automated notification alarm, prompting stakeholders to verify and investigate the source of the error, as illustrated in [Figure 10](#). This auto-verification tool serves as a leading indicator to prevent mismeasurement and plays a critical role in maintaining the accuracy and reliability of the metering system.

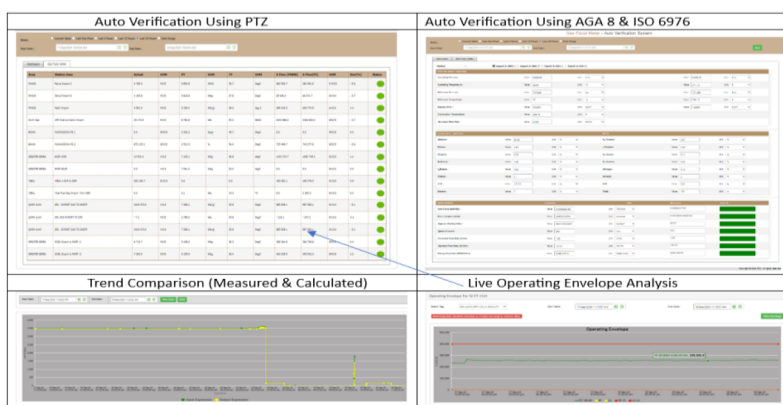


Figure 10—Auto verification Tool

The PMMS application is integrated with several key software platforms to improve data reliability, streamline operations, and enhance decision-making. These integrations ensure timely maintenance actions, comprehensive visibility, and effective monitoring. The integrated systems include:

- SAP application: Integrated to track and ensure that preventive maintenance activities for the metering systems are performed on schedule. This helps maintain meter accuracy, prevent premature failures, and reduce the risk of unexpected downtime. The system uses a color-coded alert mechanism to indicate overdue tasks, enabling quick and proactive responses.
- Asset Management System (AMS): Provides a centralized view of all PMMS-related parameters, allowing users to easily monitor, analyze, and manage asset maintenance data Nibras application: To support the system
- Process Information (PI): Enables real-time data acquisition and historical data analysis. This integration allows for performance tracking and trend analysis to support predictive maintenance and operational optimization
- Nibras: Supports overall system functionality by acting as a middleware or support layer, ensuring seamless communication and data flow between integrated applications
- Energy component (EC): Facilitates the alignment of maintenance activities with energy performance tracking and reporting, ensuring that maintenance contributes to energy efficiency goals.
- Distributed Control System (DCS): Integration with DCS provides live operational data, which is crucial for condition-based maintenance decisions and immediate system feedback.

Key features and Benefits

The Production Measurement Management System (PMMS) is a digital platform designed to enhance the accuracy, reliability, and efficiency of metering operations. By integrating with multiple enterprise applications and automating key processes, PMMS delivers tangible operational benefits and supports the continuous improvement of production measurement activities as following:

- Enhanced Visibility: PMMS provides Production Engineers with centralized and real-time access to metering data, enabling the identification of trends, anomalies, and performance issues. This visibility supports proactive monitoring and data-driven troubleshooting.
- Improved Decision-Making: Through the contextualization and correlation of data across systems (e.g., DCS, PI, Nibras, EC), PMMS equips stakeholders with accurate, actionable insights. This leads to smarter operational decisions and improved performance outcomes.

- **Reduced Downtime:** Real-time monitoring and automated alert mechanisms enable early detection of metering deviations or failures. Prompt intervention minimizes unplanned downtime and preserves production continuity.
- **Optimized Resource Utilization:** PMMS streamlines workflows and consolidates data access, significantly reducing manual efforts, duplication of work, and data entry errors. This efficiency results in better use of technical and operational resources.

The PMMS platform delivers a wide range of functionalities that significantly enhance the management of production measurement systems. It enables the comprehensive identification and registration of all metering tags across integrated applications, including DCS, PI, Nibras, EC, and PMMS itself. Through effective data management, verification, and assurance mechanisms, the platform ensures the reliability and accuracy of production measurement data. PMMS supports end-to-end lifecycle management of metering systems and provides continuous monitoring of the health and performance of various meter types. A centralized interface allows users to access multiple corporate databases seamlessly, while online tools facilitate flow compensation and standardization verification. The system generates automated alerts for BS&W (Basic Sediment & Water) threshold breaches and flare limit exceedances at the station level, helping to maintain operational integrity. Real-time production dashboards offer live asset monitoring, and enhanced tracking tools improve utilization visibility for well testing equipment such as Test Separators, MPFMs, and Net Oil Computers. PMMS also supports performance validation of well testing methods, automatic updating of fiscal meter factor control charts, and includes flow homogeneity verification features. Online workflows streamline the tracking and management of action items. The platform is fully integrated with enterprise systems such as DCS, PI, Nibras, EC, AMS, and SAP, ensuring seamless data exchange. Furthermore, it enables compliance monitoring for maintenance activities across various metering classes and provides automated verification and calculation of standard conditions for both liquid and gas measurements.

Conclusion

PMMS plays a critical role in transforming metering operations through automation, integration, and digital intelligence. It empowers technical teams to manage production measurement with greater confidence, precision, and efficiency—contributing to safer operations, improved compliance, and reduced operational costs.

References

- Albusafi, Adil, Albalush, Farah, Alshuily, Mohammed, Albalushi, Najim, Alrahali, Maadh, Alhinai, Ashraf, Alsahlani, Ali, Alsalmi, Houd, Alkiumi, Yousuf, Ram, Nayakam, Muhammed, Jithin, and Somaya Maghdari. "Power of 6 Sigma to Overcome Chronic Business Pain Area." Paper presented at the Middle East Oil, Gas and Geosciences Show, Manama, Bahrain, February 2023. doi: <https://doi.org/10.2118/213674-MS>